AEIC-MD: mechanism-mediated multi-agent deliberation

A public leader rule changes what followers choose to do before reports are aggregated.

FOLLOWER MICRO-ARCHITECTURE

FOLLOWER AGENT 1

Observe	task x · offer m · history h_i^t
Choose action	LOW / AEIC / VERIFY / IMITATE
Execute	action-specific prompt + resource
Report	AEIC = {A, E, I, C}

FOLLOWER AGENT 2

Observe	task x · offer m · history h_i^t
Choose action	LOW / AEIC / VERIFY / IMITATE
Execute	action-specific prompt + resource
Report	AEIC = {A, E, I, C}

FOLLOWER AGENT 3

Observe	task x · offer m · history h_i^t
Choose action	LOW / AEIC / VERIFY / IMITATE
Execute	action-specific prompt + resource
Report	AEIC = {A, E, I, C}

offer m

offer m

a_i, r_i

offer m

a_i, r_i

h_i^{t+1}

LEADER–FOLLOWER MECHANISM

1 Public contract
credit rule g_m · costs $c_m(a)$
information policy · verification budget

2 Action enforcement
declared action selects the actual
prompt, token limit, and resource path

3 Observable settlement
AEIC completeness · valid conclusion
credit − cost → realized utility

4 Persistent state
offer · action trace · raw report
reputation ρ_i · tokens · latency

Round controller
synchronous rounds · rollback on failure

reports + policy

COLLECTIVE DELIBERATION + AUDIT

Information environments
independent · evidence update · social only
copy pressure · counterevidence

Synchronous interaction
round t reads only permitted reports from t−1
same-round agents remain mutually hidden

Collective decision
explicit C fields → round-3 majority vote

Behavior
effort rate
actions
stance flips
novelty

Decision
accuracy
corrected
harmed
domain effect

Cost / audit
tokens
latency
failures
hashes

IDENTIFICATION

same question · seed · panel · rounds · information; only the public mechanism changes